

B.Tech. Degree V Semester Supplementary Examination June 2024CE/CS/EC/IT/ME/EE/SE 19-200-0501 NUMERICAL AND STATISTICAL METHODS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Solve algebraic and transcendental equations by numerical methods.
 CO2: Solve numerical differentiation and integration problems.
 CO3: Compute the mean and variance of a probability distribution including the binomial distribution.
 CO4: Test hypotheses on data.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24) Marks BL CO PO

- I. (a) Prove that with usual notations

(i) $hD = -\log(1 - \nabla)$

3 L1, L3 1 1

(ii) $\Delta = \frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{\delta^2}{4}}$

- (b) Given the values

3 L3 1 1

X:	5	7	11	13	17
f(x):	150	392	1452	2366	5202

Evaluate f(9) using Newton's divided difference formula.

- (c) Evaluate
- $\int_0^1 \frac{dx}{1+x^2}$
- using Simpson's
- $\frac{1}{3}$
- rd rule taking
- $h = \frac{1}{4}$
- .

3 L1, L3 1 3

- (d) Use Euler modify method to find y for x = 0.1 given

3 L3 2 2

$\frac{dy}{dx} = \frac{y-x}{y+x}, y(0) = 1.$

- (e) A random variable X has the probability distribution

x:	0	1	2	3
p(x):	$\frac{k}{2}$	$\frac{k}{3}$	$\frac{k+1}{3}$	$\frac{2k-1}{6}$

Find k, Also find mean and variance of the distribution.

- (f) A product is 0.5% defective and is packed in cartons of 100. What % contains not more than 3 defectives?

3 L3, L4 3 3

- (g) Define:

3 L1 4 1

- (i) parameter and statistic
 (ii) critical region
 (iii) test statistic.

- (h) A random sample of 900 members has a mean 3.4 cm. Can it be reasonably regarded as a sample from a large population of mean 3.2 cm and SD 2.3 cm.

3 L4 4 4

(P.T.O.)

PART B

(4 × 12 = 48)

Marks BL CO PO

- II. (a) Find the root of the equation $xe^x = \cos x$ using Regula-Falsi method correct to 4 decimal places.

6

L3

1

2

- (b) Solve the system of equation by Gauss Seidel method:

6

L3

1

2

(i) $5x + 2y + z = 12$

(ii) $x + 4y + 2z = 15$

(iii) $x + 2y + 5z = 20$

OR

- III. (a) Using Lagrange's interpolation method find the value of $f(x)$ at $x = 5$ given the values.

6

L3

1

2

x:	1	3	4	6
fx:	3	9	30	132

- (b) Given $f(40) = 184$, $f(50) = 204$, $f(60) = 226$, $f(70) = 250$, $f(80) = 276$, $f(90) = 304$. Find $f(38)$ and $f(85)$ using suitable interpolation formula.

6

L3

1

3

- IV. (a) Using Runge-Kutta method, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at

6

L3

2

2

$x = 0.2$.

- (b) A slider in a machine moves along a fixed straight rod. Its distance x cm along the rod is given below for various values of the time t seconds. Find the velocity of the slider and its acceleration when $t = 0.3$ seconds.

6

L2,

2

3

L3

t:	0	0.1	0.2	0.3	0.4	0.5	0.6
x:	30.13	31.62	32.87	33.64	33.95	33.81	33.24

OR

- V. (a) Apply Euler modified method find $y(0.2)$ and $y(0.4)$ given $y(0) = 0$, $y' = y + e^x$.

6

L2,

2

2

L3

- (b) The velocity 'v' (km/min) of a moped which starts from rests, is given at fixed intervals of time 't' (min) as follows:

6

L3

2

4

t:	2	4	6	8	10	12	14	16	18	20
x:	10	18	25	29	32	20	11	5	2	0

Evaluate approximately the distance covered in 20 minutes.

6

L1

3

1

- VI. (a) Derive mean and variance of Poisson Distribution.
(b) Phone calls arrive at the rate of 48 per hour at the reservation desk for Indian Airlines.

6

L4

3

3

- (i) Compute the probability of receiving three calls in a 5 minutes interval of time.

- (ii) Compute the probability of receiving exactly 10 calls in 15 minutes.

- (iii) Suppose no calls are currently on hold if the agent takes 5 minutes to complete the current call, how many do you expect to be waiting by that time? What is the probability that none will be waiting?

OR

6

L3

3

2

- VII. (a) Fit a parabola $y = a + bx + cx^2$ to the following data:

x:	1	3	5	7	9
y:	3	13	31	57	91

6

L3,

3

3

- (b) In a Normal distribution 17% of the items are below 30 and 17% of the item above 60. Find the mean and standard deviation.

(Continued)

BTS-V(S)-06-24-2683

		Marks	BL	CO	PO
VIII.	(a) It is claimed that sample of 100 tyres with mean life of 15269 km. is drawn from a population of tyres which has a mean life of 15200 km and standard deviation of 1248 km. Test the validity of the claim.	6	L4, L5	4	4
2 = 55 Accp	(b) The average marks in Mathematics of a sample of 100 students was 51 with a standard deviation of 6 marks. Could this have been a random sample from a population with average marks 50?	6	L4	4	4
1.66 17cc					
OR					
IX.	(a) A manufacturer claims that only 4% of his products supplied by him are defective. A random sample of 600 products contained 36 defectives. Test the claim of the manufacturer.	6	L4	4	3
	(b) A random sample of 200 villages was taken from district A and average population per village was 485 with standard deviation 50. Another random sample of 250 villages from the same district gave an average population of 510 per village with standard deviation of 40. Is the difference between the averages of two sample statistically significant?	6	L4	4	4

Blooms's Taxonomy Levels

L1 - 10%, L2 - 5%, L3 - 53.75%, L4 - 28.75%, L5 - 2.5%.

$$\sigma = 1 - \epsilon$$

B.Tech. Degree V Semester Regular/Supplementary Examination November 2023

CSE/EC/IT/CE/EE/ME/SE 19-200-0501 NUMERICAL AND STATISTICAL METHODS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Solve algebraic and transcendental equations by numerical methods.
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 CO4: Test hypotheses on data.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24)

Marks
3BL
L3CO
1PO
3

- I. (a) Find the missing term in the table:

X:	2	3	4	5	6
Y:	45.0	49.2	54.1		67.4

- (b) By Regula-Falsi method, find a real root of
- $x^2 - 9x + 1 = 0$
- .

- (c) Prove that
- $h^2 f''(a) = \Delta^2 f(a) - \Delta^3 f(a) + \left(\frac{11}{12}\right) \Delta^4 f(a) - \dots$

- (d) Apply Euler's method to find
- y
- for
- $x = 0.1$
- . Given
- $\frac{dy}{dx} = \frac{y-x}{y+x}$
- , with

initial condition $y(0) = 1$.

- (e) If random variable has the following probability distribution

X:	-2	-1	0	1	2	3
P(X):	0.1	k	0.2	2k	0.3	k

Find the value of k , mean and variance

- (f) In normal distribution 31% of the items are under 45 and 8% are over 64. Find mean and variance of the distribution.

- (g) Define the following terms:

- (i) Parameter and statistic
 (ii) Type I error and Type II error
 (iii) Significance level.

- (h) The average score in mathematics of a sample of 100 students was 51 with a standard deviation of 6 points. Could this have been a random sample from a population with average score 50?

(P.T.O.)

PART B

(4 × 12 = 48)

Marks BL CO PO

- II. (a) Find a real root of $x^3 - 2x - 5 = 0$ near $x = 2$ correct to 3 decimal places by Regula-Falsi method. 6 L3 1 2
- (b) Using Newton's forward interpolation formula, find $y = f(8)$ from the following table. 6 L4 1 4

x:	0	5	10	15	20	25
y:	7	11	14	18	24	32

OR

- III. (a) From the following table of half yearly premium for policies maturing at different ages, estimate the premium for policies maturing at age 46. 6 L4 1 4

Age (x):	45	50	55	60	65
Premium (y):	1150	960	830	740	680

- (b) Using Newton's divided difference formula, find the values of $f(2)$ and $f(15)$ given the following table: 6 L4 1 3

x:	4	5	7	10	11	13
f(x):	48	100	294	900	1210	2028

- IV. (a) Compute the value of $\int_0^1 \frac{dx}{1+x^2}$, using: 6 L4 2 4

(i) Simpson's $\frac{1}{3}$ rule, $h = \frac{1}{4}$.

(ii) Simpson's $\frac{3}{8}$ rule, $h = \frac{1}{6}$.

- (b) Apply the fourth order Runge-Kutta method to find $y(0.2)$, given $y' = x + y$, $y(0) = 1$. 6 L4 2 3

OR

- V. (a) By modified Euler's method, find the value of y when $x = 0.2$, given that $y' = x + y$, $y(0) = 1$. 6 L4 2 4

- (b) Using the Runge-Kutta method find $y(0.2)$ if $y' = x + y^2$, given that $y(0) = 1$, in steps of 0.1. 6 L4 2 4

- VI. (a) The probability that any ship of a company being destroyed on a certain voyage is 0.02. The company owns 6 ships for the voyage. What is the probability of: 6 L4 2 4

(i) losing one ship?

(ii) losing at the most two ships?

(iii) losing none?

- (b) Fit a power curve to the following data: 6 L4 2 3

x:	1	2	3	4	5	6
y:	2.98	4.26	5.21	6.1	6.8	7.5

OR

- VII. (a) Out of 800 families with 4 children each, how many families would be expected to have (i) 2 boys and 2 girls (ii) at least one boy (iii) no girl (iv) at most two girls? Assume equal probabilities for boys and girls. 6 L4 2 4

- (b) In a competitive examination, 5000 students have appeared for a paper in Engineering Mathematics. Their average marks was 62 and standard deviation 12. If there are only 100 vacancies, find the minimum marks that one should score in order to get selected. 6 L4 3 4

BTS-V(R/S)-11-23-2050

Marks BL CO PO

- VIII. (a) Intelligent test were given to two groups of boys and girls. Given the following data:

6 L4 4 4

	Mean	S.D.	Size
Girls	75	8	60
Boys	73	10	100

Test the difference between mean scores is significant.

- (b) Two random samples reveal the following data:

6 L4 4 4

	Size	Mean	Variance
Sample I	16	440	40
Sample II	25	460	42

Test whether the samples came from the same normal population.

OR

- IX. (a) Two independent random sample of size $n_1 = 10$, $n_2 = 7$ when observed to have sample variances,

6 L4 4 4

$$S_1^2 = 16, S_2^2 = 3 \text{ using LOS @ 1\%}$$

$$\text{Test } H_0: \sigma_1^2 = \sigma_2^2 \text{ Vs } H_1: \sigma_1^2 \neq \sigma_2^2$$

- (b) The standard deviation of 20 observations from a normal population was found to be 5. Examine whether the sample was taken from a population with standard deviation 5.3.

6 L4 4 4

Bloom's Taxonomy Levels

L1 = 10%, L2 = 35.83%, L3 = 44.17%, L4 = 10%.

B.Tech. Degree V Semester Regular/Supplementary Examination January 2023

CS/EC/IT/CE/EE/ME/SE 19-200-0501 NUMERICAL AND STATISTICAL METHODS

(2019 Scheme)

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PART A

(Answer ALL questions)

(8 × 3 = 24)

- | | Marks | BL | CO | PO | | | | | | | | | | | | |
|--|-------|-----|-----|-----|------|---|------|-----|---|---|---|----|--|--|--|--|
| I. (a) Find the missing term in the following table. | 3 | L2 | 1 | 1.1 | | | | | | | | | | | | |
| <table border="1" style="display: inline-table; vertical-align: middle;"> <tr> <td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr> <tr> <td>f(x)</td><td>1</td><td>3</td><td>9</td><td>-</td><td>81</td></tr> </table> | x | 0 | 1 | 2 | 3 | 4 | f(x) | 1 | 3 | 9 | - | 81 | | | | |
| x | 0 | 1 | 2 | 3 | 4 | | | | | | | | | | | |
| f(x) | 1 | 3 | 9 | - | 81 | | | | | | | | | | | |
| (b) A devotee of Newton-Raphson method used the method to solve the equation $x^{100} = 0$ using the initial estimate $x_0 = 0.1$. Calculate the next three estimate using Newton method. | 3 | L3 | 1 | 1.1 | | | | | | | | | | | | |
| (c) The f(x) is given by | 3 | L3 | 2 | 1.1 | | | | | | | | | | | | |
| <table border="1" style="display: inline-table; vertical-align: middle;"> <tr> <td>x</td><td>0</td><td>0.5</td><td>1</td></tr> <tr> <td>f(x)</td><td>1</td><td>0.8</td><td>0.5</td></tr> </table> | x | 0 | 0.5 | 1 | f(x) | 1 | 0.8 | 0.5 | | | | | | | | |
| x | 0 | 0.5 | 1 | | | | | | | | | | | | | |
| f(x) | 1 | 0.8 | 0.5 | | | | | | | | | | | | | |
| Then using Trapezoidal rule what is the value of $\int_0^1 f(x)dx$? | | | | | | | | | | | | | | | | |
| (d) Using Runge-kutta method of order 4 what is the value of $y(0.1)$ for $y' = x - 2y$, $y(0) = 1$ taking $h = 0.1$. | 3 | L2 | 2 | 1.1 | | | | | | | | | | | | |
| (e) Given. | 3 | L2 | 3 | 1.1 | | | | | | | | | | | | |
| <table border="1" style="display: inline-table; vertical-align: middle;"> <tr> <td>x</td><td>0</td><td>1</td><td>2</td></tr> <tr> <td>y</td><td>0</td><td>1.1</td><td>2.1</td></tr> </table> | x | 0 | 1 | 2 | y | 0 | 1.1 | 2.1 | | | | | | | | |
| x | 0 | 1 | 2 | | | | | | | | | | | | | |
| y | 0 | 1.1 | 2.1 | | | | | | | | | | | | | |
| Then which is the best fit straight line? | | | | | | | | | | | | | | | | |
| (f) If $P(1) = P(2)$ then what is the mean of the Poisson distribution. | 3 | L2 | 3 | 1.1 | | | | | | | | | | | | |
| (g) A random sample of 400 items gives the mean 4.45 and the variance as 4. Can the samples be regarded as drawn from the normal population with mean 4? (At 5% level of significance). | 3 | L3 | 4 | 1.1 | | | | | | | | | | | | |
| (h) Derive the formula for sampling distribution of means. | 3 | L1 | 4 | 1.1 | | | | | | | | | | | | |

PART B

(4 × 12 = 48)

- | | | | | |
|---|---|----|---|-----|
| II. (a) Using regular-falsi method compute the real root of the following equations correct to three decimal places.
$2x - \log x = 7$ | 6 | L3 | 1 | 1.1 |
| (b) Using Newton's divided difference interpolation find the polynomial of the given data | 6 | L3 | 1 | 1.1 |

x	-1	0	1	3
f(x)	2	1	0	-1

OR

(P.T.O.)

		Marks	BL	CO	PO														
III.	(a) If $y(1) = -3, y(3) = 9, y(4) = 30, y(6) = 132$ find the Lagrange's interpolation polynomial that takes the same values as y at the given points.	6	L3	1	1.1														
	(b) Solve the following equation by Gauss-Seidel method $2x + y + 6z = 9, 8x + 3y + 2z = 13, x + 5y + z = 7$.	6	L3	1	1.1														
IV.	(a) Find the $y'(0)$ and $y''(0)$ from the following table	6	L5	2	1.1														
	<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>4</td><td>8</td><td>15</td><td>7</td><td>6</td><td>2</td></tr></table>	x	0	1	2	3	4	5	y	4	8	15	7	6	2				
x	0	1	2	3	4	5													
y	4	8	15	7	6	2													
	(b) Apply Euler's method to solve $y' = x + y, y(0) = 0$ choosing the step length = 0.2 (carry out 6 steps).	6	L5	2	1.1														
OR																			
V.	(a) Solve $y' = 3x + y^2, y(0) = 1$. Using Taylor's series method and compute $y(0.1)$.	6	L4	2	1.1														
	(b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using:	6	L4	2	1.1														
	(i) trapezoidal rule																		
	(ii) Simpson's 1/3 rd rule																		
	(iii) Simpson's 3/8 th rule.																		
VI.	(a) From an urn containing 3 red and 2 white balls a man is to draw 2 balls at random without replacement, being promised ₹20 for each red ball he draws and ₹10 for each white one. Find his expectation.	6	L4	3	1.1														
	(b) Fit a binomial distribution for the following data.	6	L4	3	1.1														
	<table><tr><td>X</td><td>0</td><td>1</td><td>3</td><td>4</td></tr><tr><td>y</td><td>28</td><td>62</td><td>10</td><td>4</td></tr></table>	X	0	1	3	4	y	28	62	10	4								
X	0	1	3	4															
y	28	62	10	4															
OR																			
VII.	(a) In a normal distribution 31% of the items are under 45 and 81% are over 64. Find the mean and standard deviation of the distribution.	6	L5	3	1.1														
	(b) By the method of least squares find the straight line that best fits the following data.	6	L5	3	1.1														
	<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>14</td><td>27</td><td>40</td><td>55</td><td>68</td></tr></table>	x	1	2	3	4	5	y	14	27	40	55	68						
x	1	2	3	4	5														
y	14	27	40	55	68														
VIII.	(a) A die was thrown 60 times and the following frequency distribution was observed.	6	L4	4	1.1														
	<table><tr><td>Faces</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>f_0</td><td>15</td><td>6</td><td>4</td><td>7</td><td>11</td><td>17</td></tr></table>	Faces	1	2	3	4	5	6	f_0	15	6	4	7	11	17				
Faces	1	2	3	4	5	6													
f_0	15	6	4	7	11	17													
	Test whether the die is unbiased?																		
	(b) Two samples of 9 and 7 individuals have variances 4.8 and 9.6 respectively. Is the variance 9.6 significantly greater than the variance 4.6?	6	L4	4	1.1														
OR																			
IX.	(a) A machine produces 16 imperfect articles in a sample of 500. After machine is overhauled, it produces 3 imperfect articles in a batch of 100. Has the machine been improved.	6	L4	4	1.1														
	(b) In a sample of 500 people from a state 280 take tea and rest take coffee. Can we assume that tea and coffee are equally popular in the state at 5% level of significance?	6	L4	4	1.1														

Blooms's Taxonomy Levels

L1 - 5%, L2 - 17%, L3 - 29%, L4 - 33%, L5 - 16%.
